PLOT for Magnetic Correction using Driving in Circles Dataset:

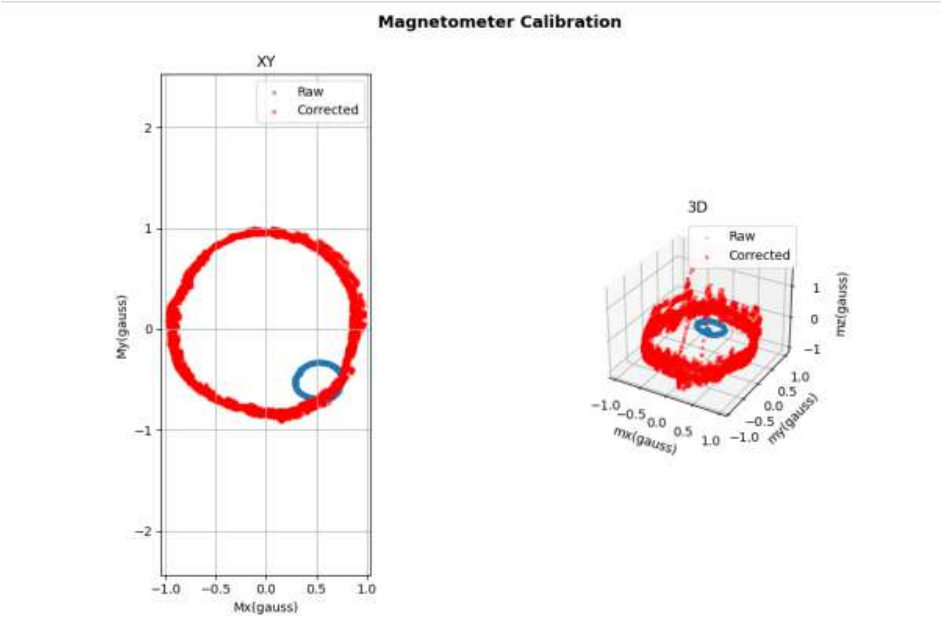


Fig1: Magnetometer data(X and Y axes), Raw vs corrected for driving_in_circles dataset

From this data, we get the Hard-iron offsets to be:

A = [[4.18840892 0.0252338 1.52507168]

[0.0252338 4.95039529 1.08457237]

[1.52507168 1.08457237 12.61154806]]

And the soft-iron offsets are:

B = [0.514558966 -0.53171244 0.81041767]

PLOTS FOR DRIVING DATASET (Boston Tour):

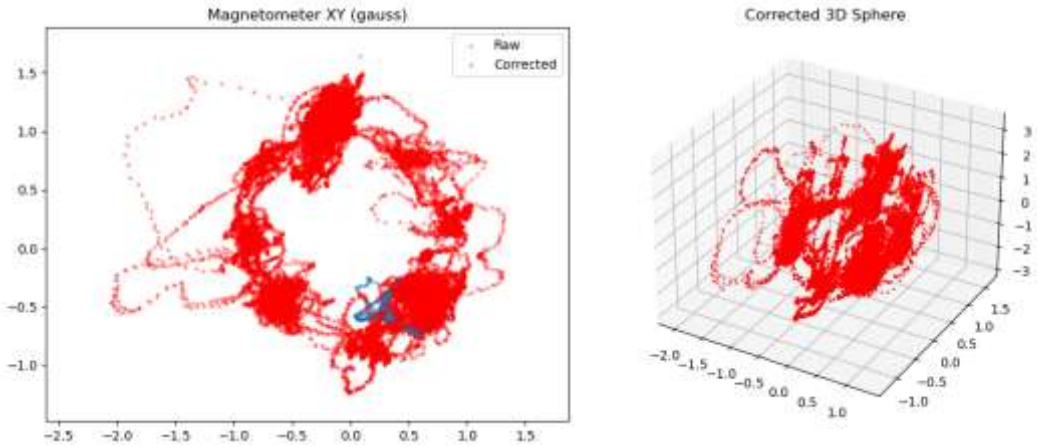


Fig2: Magnetometer data(X and Y axes), Raw vs corrected for Boston tour data

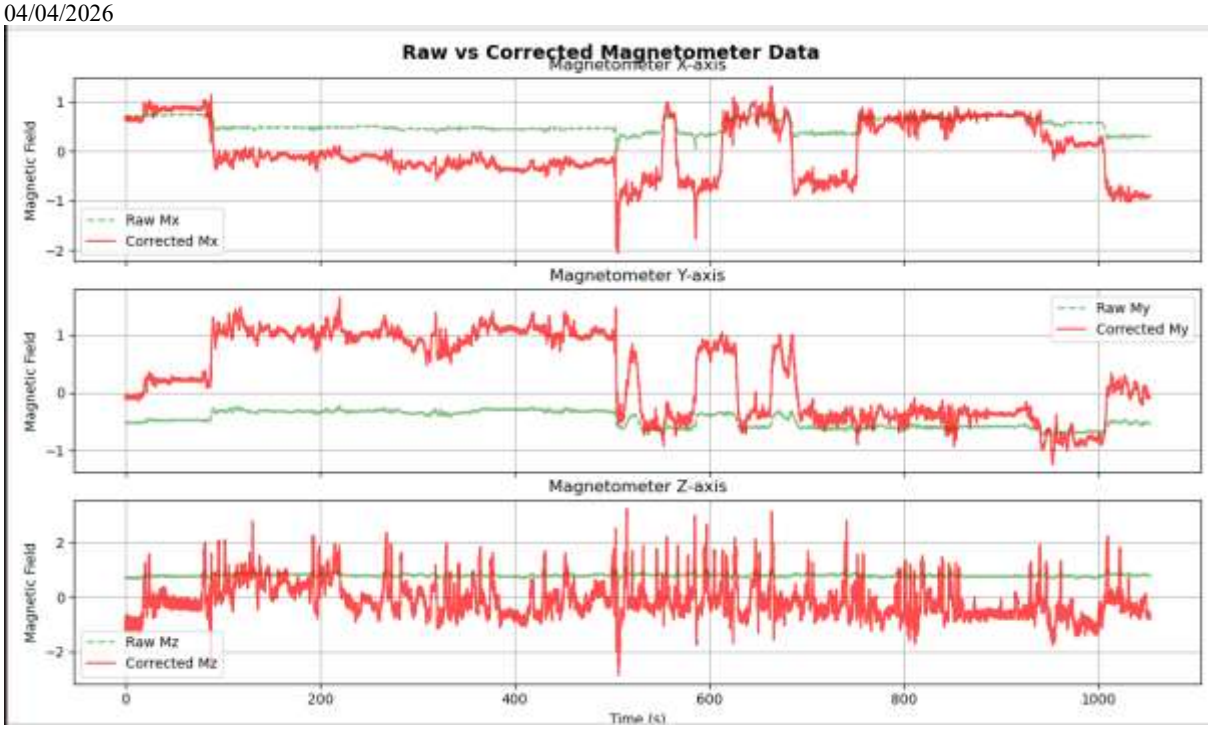


Fig3: Time Series of Raw and Corrected Magnetometer data

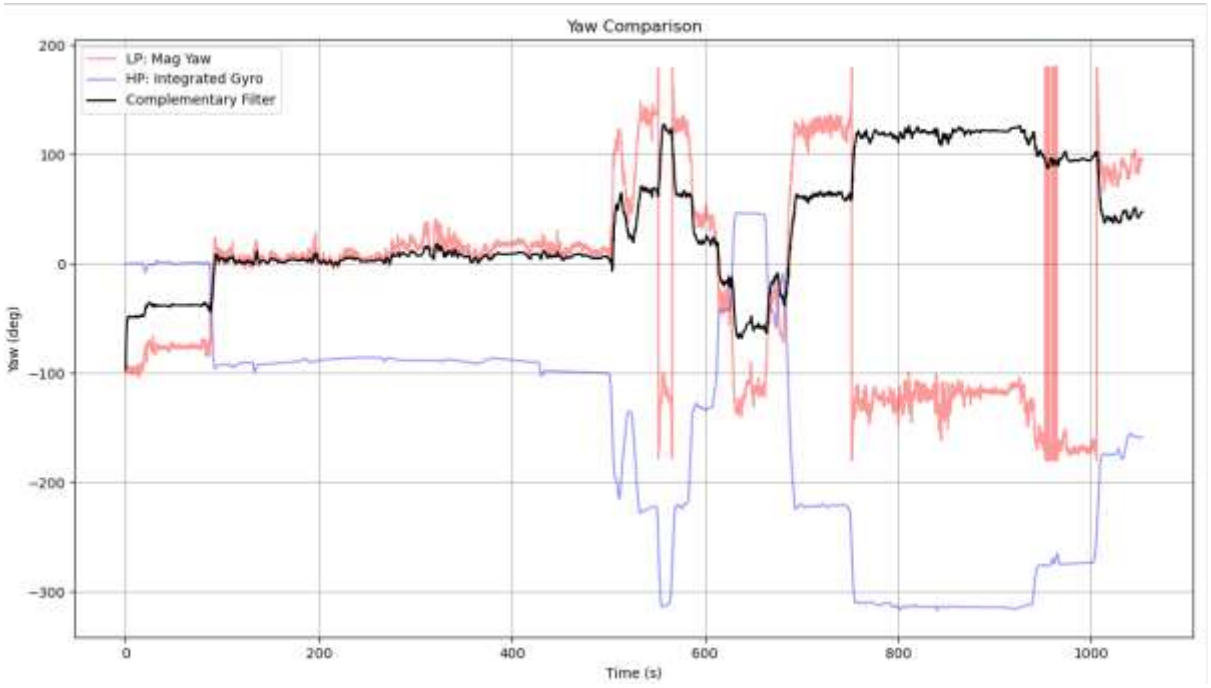


Fig4: Yaw(LP, HP, Filtered) vs Time (α=0.98)

Yaw estimated using magnetic data passed through a Low Pass filter, filters out high frequencies as observed during correction (and even though it has noticeable high frequencies it is filtered out mainly as compared to the corrected magnetometer data) and yaw estimated using gyroscope data passed through a High Pass filter filters out low frequencies in the data.

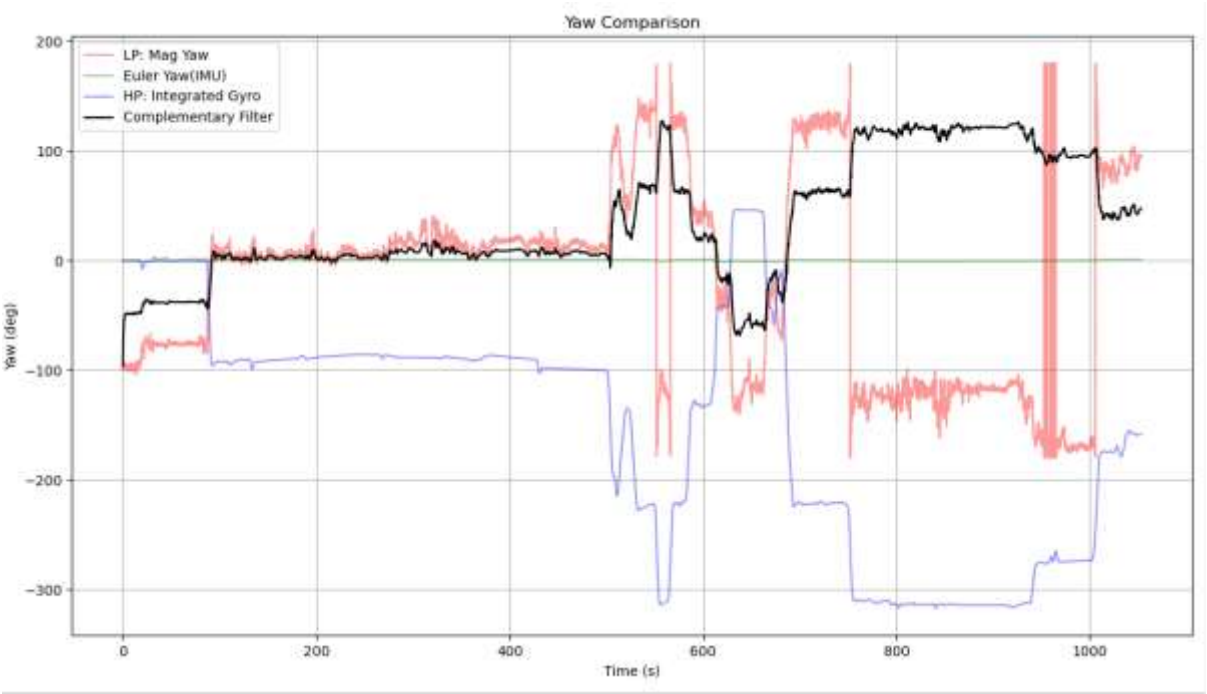


Fig5: Yaw(LP, HP, Filtered and Euler yaw) vs Time (α =0.98)



Fig6: Estimate of forward velocity using GPS data vs Time

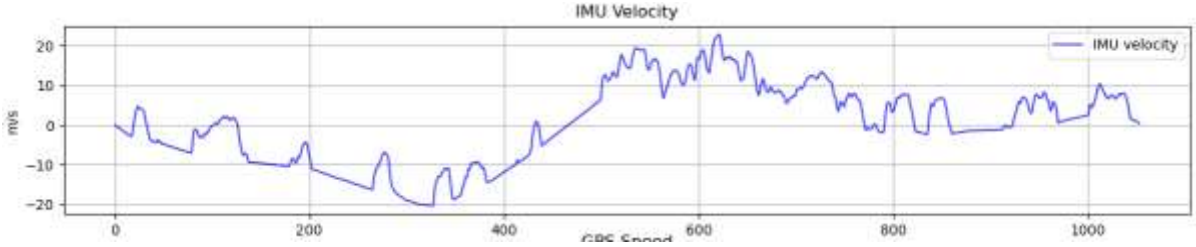


Fig7: Estimate of forward velocity using IMU(acceleration_x) data vs Time

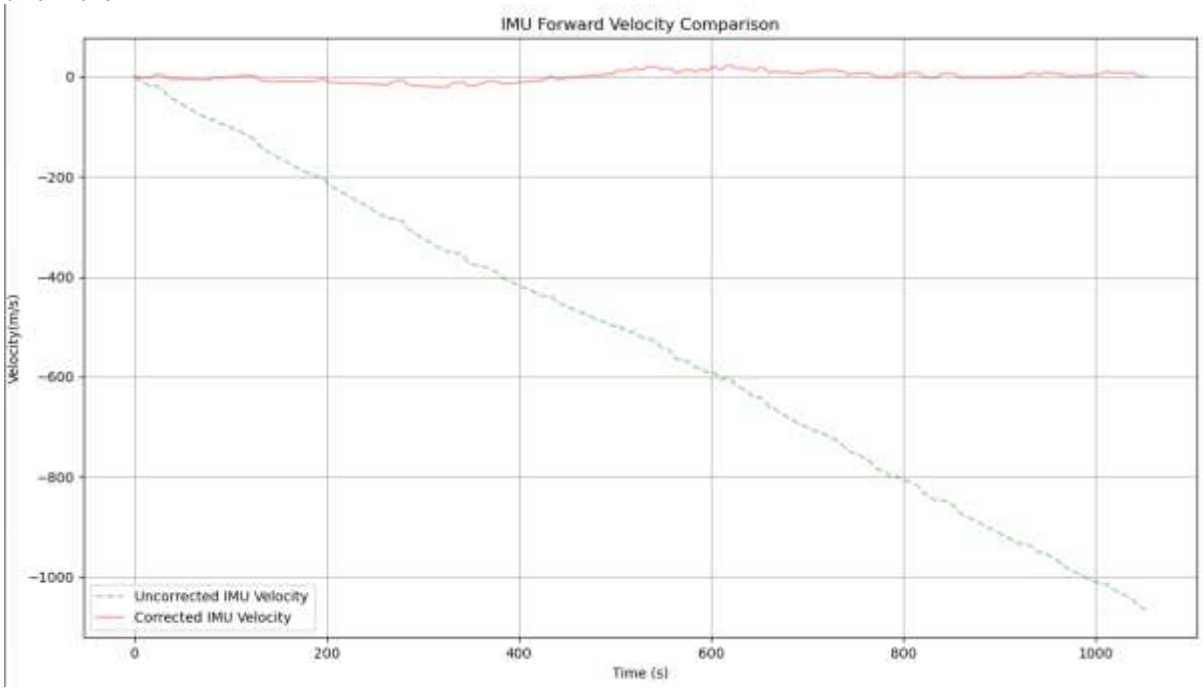


Fig8: Unbiased and biased estimate of IMU forward velocity vs Time

The uncorrected or biased acceleration computes the velocity that drifts in long term as a result of constant bias being integrated over time.



Fig9: 1-D Displacement using GPS and IMU data(drifts a lot)

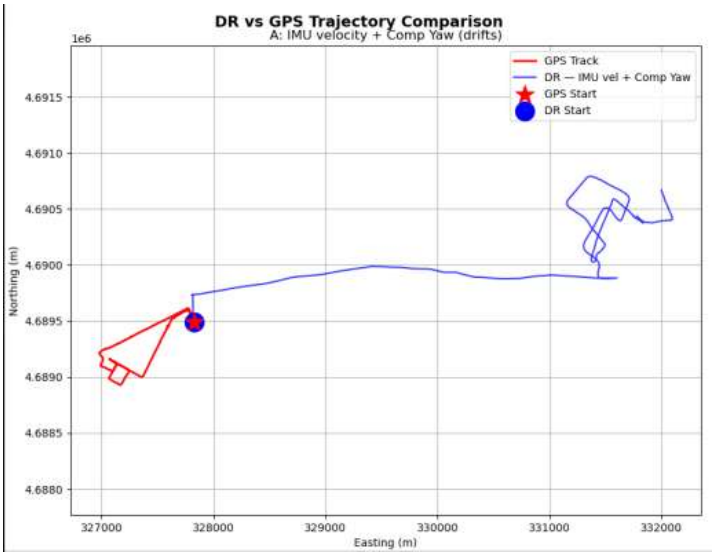


Fig10: Trajectory comparison(GPS and IMU) using no adjustments of the heading angle

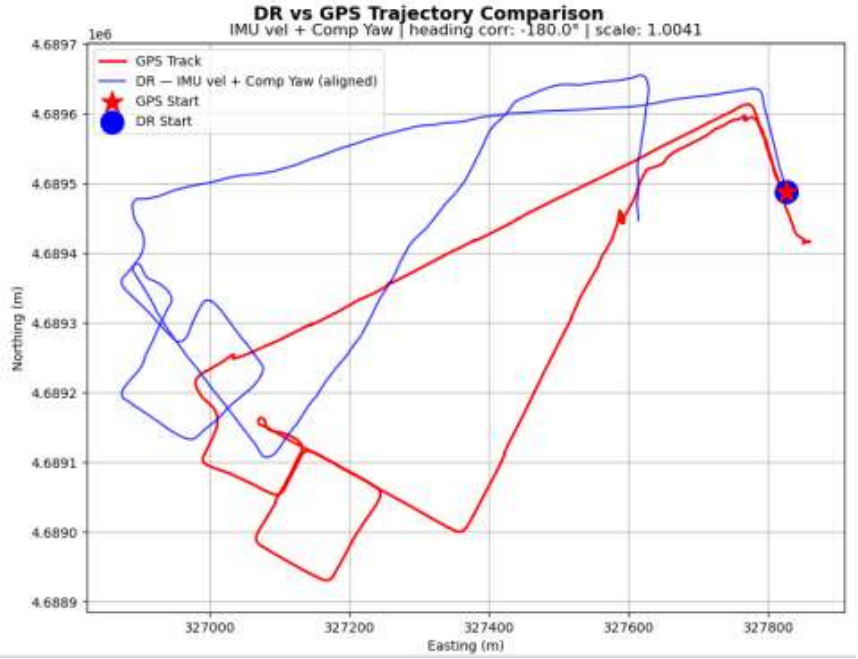
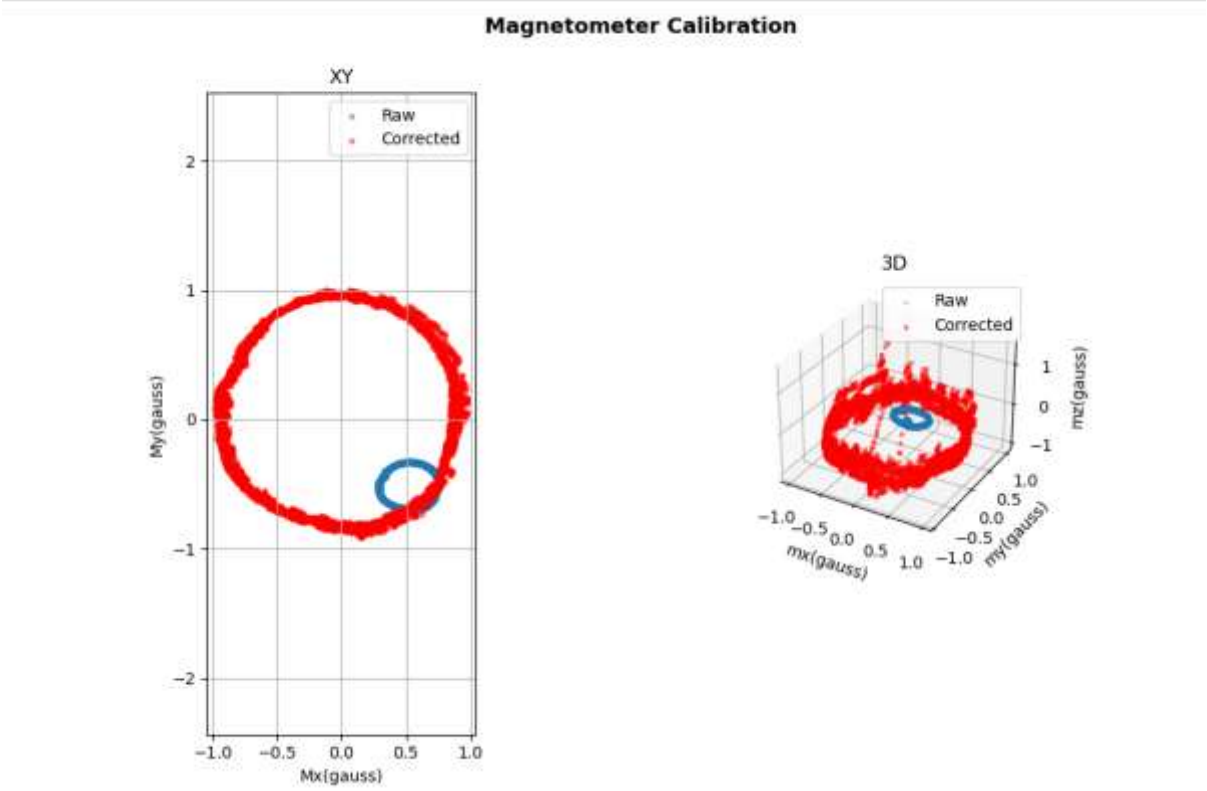


Fig11: Trajectory comparison(GPS and IMU)

using adjustments of the heading angle

Questions:

1. How did you calibrate the magnetometer from the data you collected? What were the sources of distortion present, and how do you know?
   From the driving in circles data, the magnetic data plotted for mx and my gave a circle centred around (0.5, -0.5) gauss, by fitting the raw datapoints to this circle and then using Least Squares method to compute A, the hard iron offsets and B, the soft iron offsets
   The shape of the raw data was a circle not centred around zero which means that the magnetic field around x, y axes is changing which isn't supposed to be the case in a situation where the magnetometer is supposed to measuring Earth's magnetic field. And hence there were definitely sources of distortion present which can be:
   Magnetic field due to laptops primarily or magnetic fields caused by materials like the frame of the vehicle used for data collection, cause distortion in the Earth's magnetic field in the form of soft iron distribution(negligible in this case)
   Batteries in the vehicles are another reason to cause Hard iron distortion.
   Hard iron offsets lead to cloud of points shifted away from the origin and Soft Iron offsets lead to the cloud of points being tilted or forming a ellipsoid which is not case observed in this dataset, which means that the soft iron offsets were negligible.



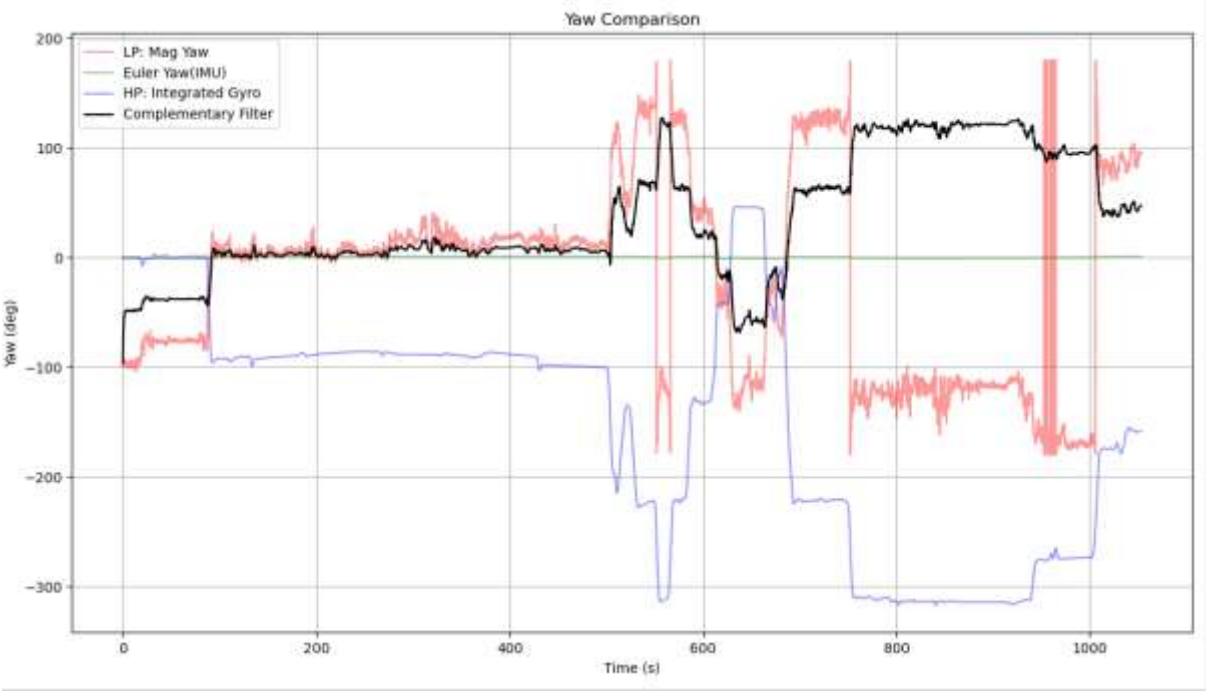
2. How did you convert raw magnetometer readings (gauss) to angles (degrees).

   I firstly corrected the magnetometer readings using Least square methods to compute the corrected magnetometer data and get the hard and soft iron offsets and then I used tangent function to convert magnetometer readings to yaw angle in radians which later on I converted to degrees. The formula used for conversion is

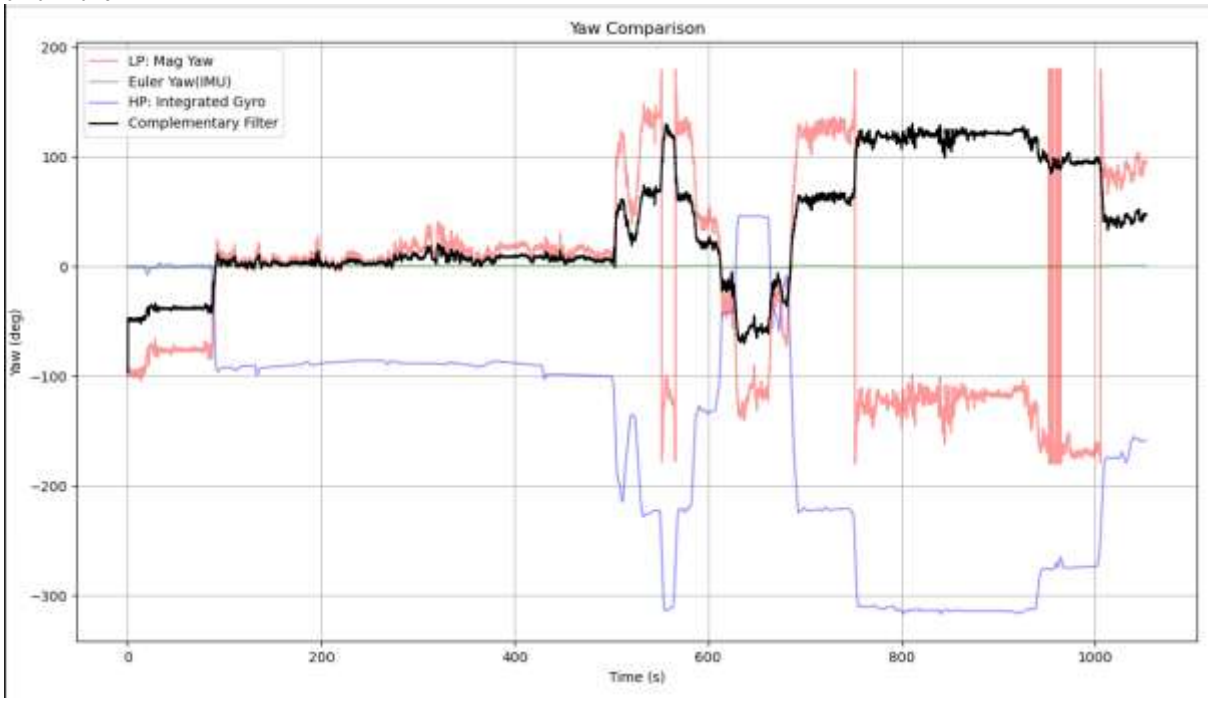
   $$\text{Yaw} = \text{atan2}\ (-my/mx)$$

3. How did you use a complementary filter to develop a combined estimate of yaw?
   I implemented a complementary filter on the yaw angle obtained through corrected magnetometer data using the formula mentioned above and the on the yaw angle obtained by integrating the angular rate about the z axis. The filter acts as a weighted sum of the yaw obtained through integrating the gyroscope data by passing it through a High-Pass filter to dampen out low frequencies and yaw obtained through the magnetometer data by passing it through Low-Pass filter to dampen out high frequencies as the magnetometer has long term stability and is not subjected to drift in long term, whereas gyroscope due to the integration to get the yaw angle is subjected to drift in long term.

4. What components of the filter were present, and what cutoff frequency(ies) did you use? Show the equation of your complementary filter.
   The complementary filter combines two components: the magnetometer yaw as the low-pass element and the integrated gyroscope as the high-pass element. With $\alpha = 0.98$, 98% of the output is derived from the gyro prediction and only 2% from the magnetometer measurement. If our IMU has sampling rate of approximately 40 Hz, the cutoff frequency is computed as $f_c = (1-\alpha) / (2\pi \cdot \alpha \cdot T) \approx 1.3020$ Hz.
   The plot including the implementation of the complementary filters shows that the fused output is much smoother than the Low pass and high pass filtered outputs.



If we consider that α=0.7 giving high influence to the gyro estimate the plot would look something like this, and the inference would be that since 30% of the weight is given to the magnetometer prediction we find that the fused signal data consist of high frequencies encoded that the Low Pass Magnetometer filter does not filter out and hence we don't get a very smooth fused signal as we would want.



The equation for the complementary filter used is:

$$\psi(t) = \alpha \cdot (\psi(t-1) + \omega_z \cdot T) + (1 - \alpha) \cdot \psi_mag$$

4. Which estimate or estimates for yaw would you trust for navigation? Why?

If the data signals were faster than the cutoff frequency computed above I would trust the gyro more since it is correct instantaneously and if the data signal were lower than the cutoff frequency computed above I would trust the magnetometer more, because it is correct in longer term than gyroscope.

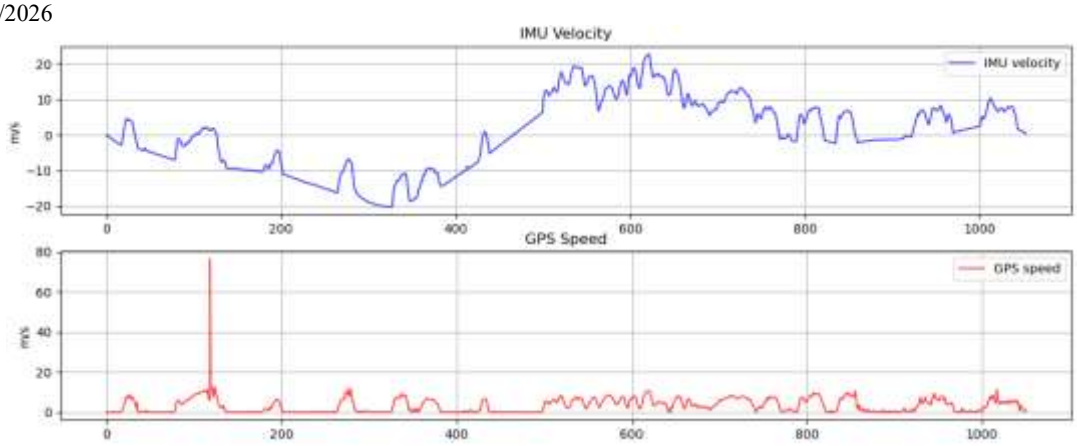
5. What adjustments did you make to the forward velocity estimate, and why?

Forward velocity estimation using acceleration along x-axis was done by debiasing the acceleration data obtained along x—axis by computing the mean and then subtracting it from each datapoint to debias it and then integrating it along the timestamps to get that, also no adjustment was done when obtaining velocity estimates from GPS sensor data

6. What discrepancies are present in the velocity estimate between accel and GPS. Why?

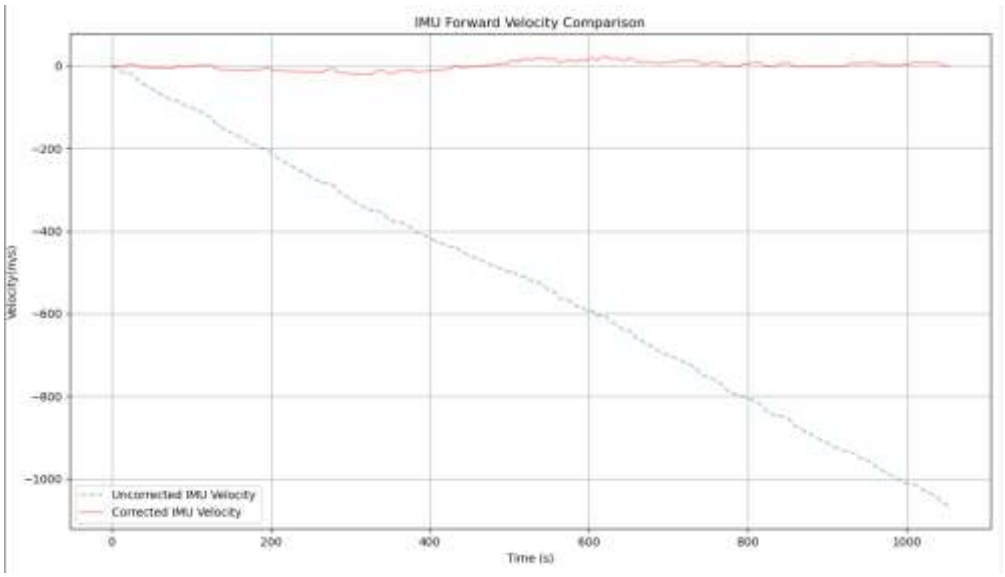
When I compute the velocity from GPS, I am estimating how much distance it travelled in some delta time.

However, when I am trying to estimate the velocity from the acceleration data obtained from the IMU by integrating it, there is high drift accumulated as the time grows because of the integration of the bias that is present that constantly adds up in integration at each time step. This causes an inaccurate estimation of velocity from the data as the time grows.

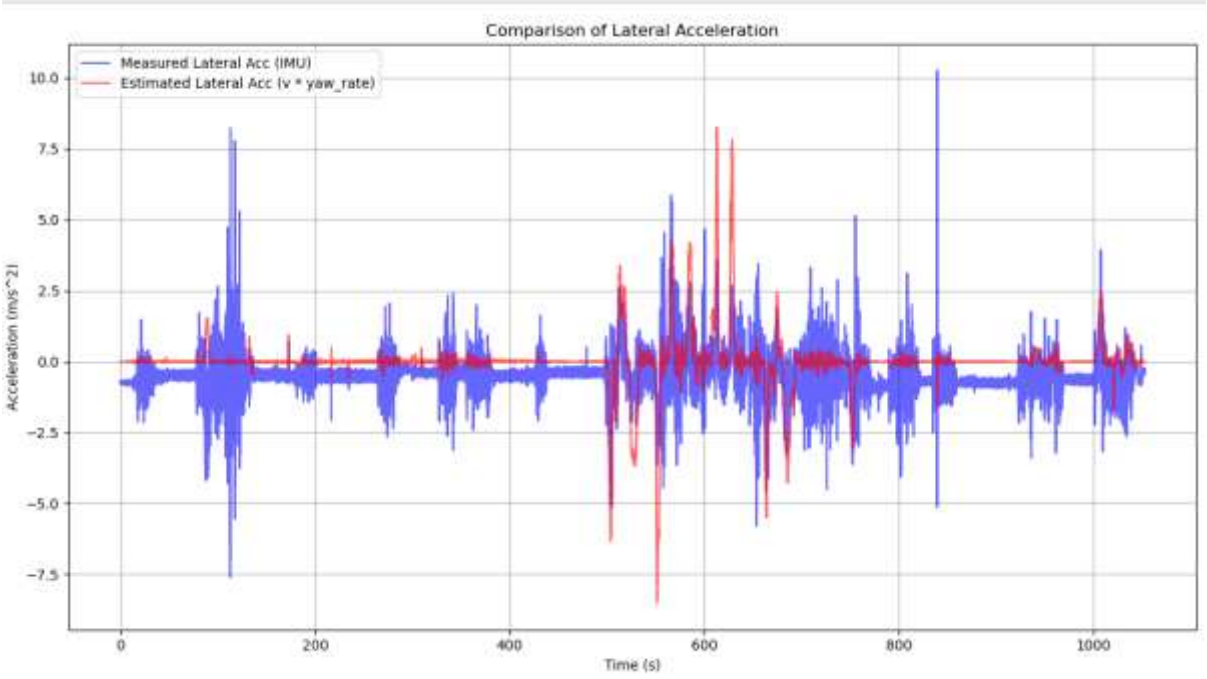


The comparison between the velocity derived from the IMU and the GPS speed highlights the fundamental trade-offs between inertial sensing and global positioning. The IMU velocity exhibits significant integration drift, where a small, constant bias in the accelerometer is integrated over time, causing the velocity estimate to wander far from the physical reality, as seen where it erroneously drops to -20 m/s despite the vehicle moving forward. In contrast, the GPS speed provides a much more stable absolute reference, staying consistently between 0 and 12 m/s, but it is susceptible to multipath errors or "GPS jumps" in urban environments, evidenced by the impossible 80 m/s spike at around 120s . While the IMU is excellent at capturing high-frequency transitions and short-term changes in motion, it lacks a long-term reference; conversely, the GPS offers long-term stability but lacks the high-frequency resolution and reliability of the IMU

he drift in the IMU data velocity estimation, if the IMU is biased then the error goes on growing at each time step causing very high error as the time grows. The uncorrected or unbiased IMU velocity estimate (dotted green line) growing unboundedly with time.

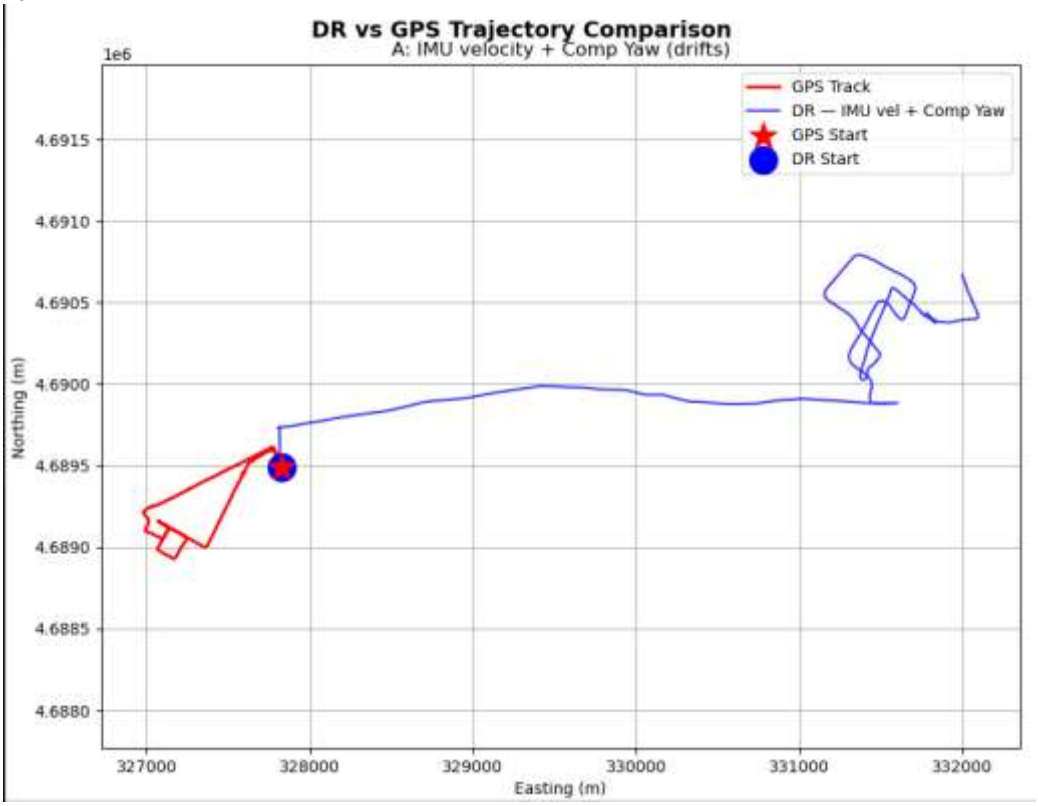


7. Compute $\omega \hat{X}$ and compare it to $\dot{y}obs$. How well do they agree? If there is a difference, what is it due to?



It is clear from the above plot that the signals are aligned when plotted with respect to relative time indicating that it captures the turning behaviour correctly. Also, the signals agree with each other, with the lateral acceleration obtained from the IMU (raw data) is observed to have more, short term high frequency variations which is more noisier whereas the computed ones are prone to less high frequency variations.
The differences are due to several factors. First, the IMU accelerometer measures all lateral forces acting on the vehicle body, including road vibrations, bumps, surface irregularities, and mechanical noise from the vehicle itself, none of which are captured by the smooth kinematic estimate ω·v. Second, the estimated lateral acceleration uses vel_imu which is derived from integrating the forward acceleration and is already known to be poorly estimated due to bias, meaning ω·v inherits those velocity errors. Third, the accelerometer also picks up gravity components if the IMU is not perfectly level, adding a constant or slowly varying offset. Finally, any misalignment between the IMU axes and the vehicle's true forward/lateral axes will cause cross-axis contamination, where forward acceleration leaks into the lateral channel and vice versa

8. Estimate the trajectory of the vehicle (xe,xn) from inertial data and compare with GPS. (adjust heading so that the first straight lines from both are oriented in the same direction). Report any scaling factor used for comparing the tracks.



Trajectory Estimation

The vehicle trajectory (xe, xn) was estimated using dead reckoning by integrating the forward velocity (derived from the IMU accelerometer) projected onto the East and North axes using the heading from the complementary filter yaw. The complementary filter fused the magnetometer yaw (low-pass) and the integrated gyroscope (high-pass) with $\alpha = 0.7$.

Heading Adjustment

The heading correction was applied by computing the initial heading from the GPS positions over some delta time t at the start of the trajectory and comparing it to the initial heading of the IMU trajectory obtained during the same time window. The correction angle was found to be -180.0°, meaning the IMU trajectory was initially pointing in exactly the opposite direction to the GPS track. This is likely due to a sign convention mismatch in the magnetometer yaw computation or a 180° mounting ambiguity in the IMU orientation.

Scaling Factor

The scaling factor applied was 1.0041, meaning the IMU arc length was only 0.41% shorter than the GPS arc length. This is remarkably close to 1, indicating that the velocity estimate from the IMU, despite its other issues, produced a total distance very close to the GPS ground truth. No significant scaling correction was needed.

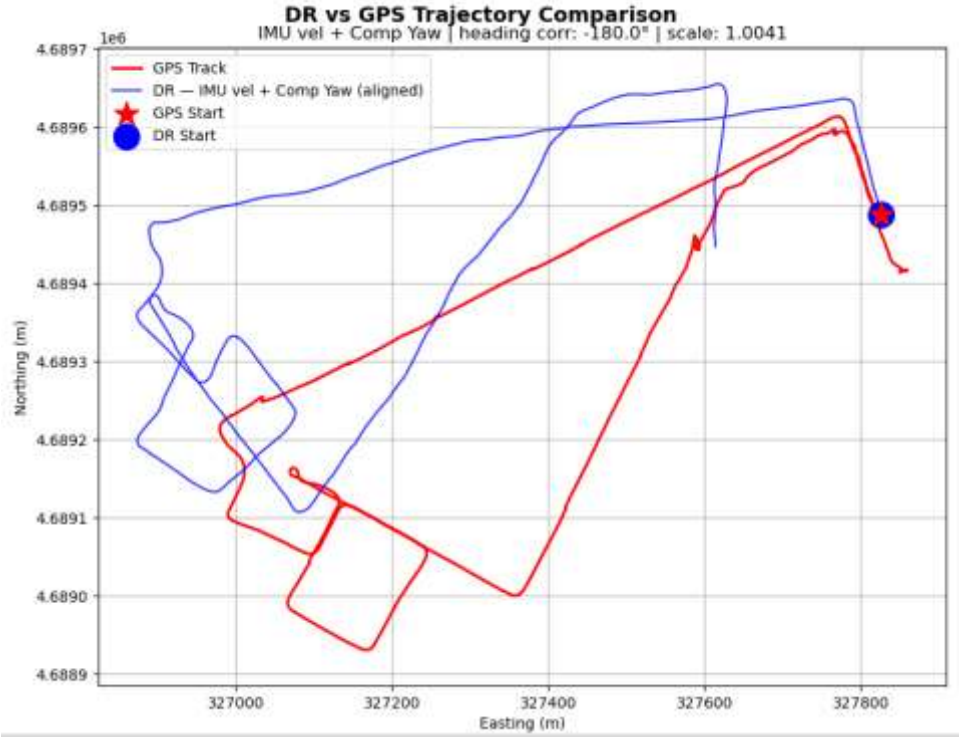
Comparison

After alignment, the two trajectories show similar overall shape and scale, both show a loop of comparable size in the same region. However, the DR trajectory (blue) does not follow the GPS track precisely. The turns occur at roughly the correct locations but with noticeable offsets, and the overall shape is expanded and rotated slightly relative to the GPS. This residual error is due to the accumulated heading errors from the complementary filter, particularly the gyro drift and magnetometer noise discussed earlier, which cause the heading estimate to be slightly wrong at every timestep, distorting the shape of the reconstructed trajectory progressively over time.

9. Given the specifications of the VectorNav, how long would you expect that it is able to navigate without a position fix? For what period of time did your GPS and IMU estimates of position match closely? (within 2 m) Did the stated performance for dead reckoning match actual measurements? Why or why not?

   Based on the VectorNav specifications, the sensor has a stated heading accuracy of around 2° and velocity accuracy sufficient for short-term dead reckoning. Using a simple error propagation estimate, a 2° heading error at a typical speed of ~5 m/s would produce a lateral position error growing at roughly 0.17 m/s, meaning the 2m threshold would be breached in about 10–12 seconds without a GPS fix.

   From your trajectory plot, the GPS and IMU estimates match closely (within approximately 2m) only for the very first few seconds near the start point, before the heading and velocity errors begin to accumulate and the trajectories visibly diverge.
   The stated dead reckoning performance did not match actual measurements. This is because the VectorNav specifications assume a well-calibrated IMU with properly removed gyro bias and accurate heading initialization, none of which were fully achieved here. The gyro drift was significant, the complementary filter yaw inherited magnetometer noise, and the velocity from accelerometer integration accumulated bias errors rapidly, all causing the DR trajectory to diverge from GPS much faster than the manufacturer specifications would suggest.




```
--- Trajectory Alignment ---
  GPS initial heading   : 0.00 deg
  INS initial heading   : -180.00 deg
  Heading correction    : -180.00 deg
  GPS arc length        : 3392.78 m
  INS arc length        : 3379.09 m
  Scaling factor applied: 1.0041  (INS underestimates distance)

--- Summary ---
  Heading correction applied : -180.00 deg
  Scaling factor applied     : 1.0041
```